2 Displacement, forces and matrix construction

We now build the system $Ax = b$ of the preceding section in three steps: the material's response, each triangle's contribution and the supports.

2.1 From displacement to internal forces

We model a thin sheet of constant thickness $t > 0$, made of one material with the same properties everywhere and in every direction. Loads act in its plane and its faces are free, so we use *plane stress*: stresses through the thickness are neglected. Loading is static, and strains and rotations are small enough for a linear model; bending and permanent deformation are excluded.

The material resists stretching and changes of angle. For displacement $\mathbf{u} = (u_x, u_y)^T$, these changes form the strain vector

$$\varepsilon(\mathbf{u}) = \begin{bmatrix} \partial u_x/\partial x \\ \partial u_y/\partial y \\ \partial u_x/\partial y + \partial u_y/\partial x \end{bmatrix}, \qquad \sigma = D\varepsilon, \qquad D = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}. \tag{4}$$

The first two entries measure stretching in the coordinate directions; the third measures the change of a right angle (engineering shear). The stress vector $\sigma = (\sigma_{xx}, \sigma_{yy}, \sigma_{xy})^T$ contains the corresponding internal forces per unit area. Young's modulus $E > 0$ controls resistance to stretching, while Poisson's ratio $-1 < \nu < 1/2$ describes the sideways response [1]. Lengths are in metres and stress and E in pascals.

2.2 From triangles to the stiffness matrix

Inside triangle e, we write displacement as $\mathbf{u}_h = \sum_{i=1}^{3} N_i \mathbf{u}_i$. Each linear weight N_i is one at vertex i and zero at the other vertices. Differentiating these weights gives a matrix B_e that converts the triangle's six vertex displacements, collected in d_e, into strain: $\varepsilon(\mathbf{u}_h) = B_e d_e$. The derivatives are constant, so strain is constant within each triangle.

We express force balance by equating internal and external work for any small test movement allowed by the supports. Substituting the triangle approximation gives its stiffness matrix

$$K_e = \int_{T_e} tB_e^T D B_e \, dA = tS_e B_e^T D B_e, \tag{5}$$

where T_e is the triangle and $S_e > 0$ its area. This 6×6 matrix converts vertex movements into internal nodal forces [3]. We add its entries to the corresponding rows and columns of K, using the displacement order $d = (u_{x,0}, u_{y,0}, \ldots, u_{x,n-1}, u_{y,n-1})^T$ for n nodes. Contributions at shared nodes are added together. Nodes without a common triangle have no direct interaction, so most entries are zero and we store K sparsely.

The square of Section 4 uses a structured mesh of squares cut along a diagonal; the other objects are meshed with Triangle [2] (Section 5). Every mesh has positive-area triangles, matching shared edges and empty holes.[1]

External forces are distributed to nodes using the same weights N_i. For example, a constant boundary force per unit area $\bar{\tau}$, called traction, gives a force per unit edge length $q = t\bar{\tau}$. An edge of length ℓ contributes $\ell q/2$ to each endpoint [3]. The load vector f is in newtons and K in N/m. Under traction loading both K and f are proportional to t, so the displacement does not depend on the thickness.

[1] We use our own assembly routine (`fem/elasticity.py`); the course-provided routine was not available.

2.3 From supports to a solvable system

The supports prescribe displacement components indexed by C; the remaining components, indexed by F, are unknown. A support can prevent movement in one direction while allowing it in the other. Taking the free rows of $Kd = f + r$ and moving known terms to the right gives

$$\underbrace{K_{FF}}_{A}\underbrace{d_F}_{x} = \underbrace{f_F - K_{FC}d_C}_{b}. \tag{6}$$

The subscripts select rows and columns, respectively. This recovers $Ax = b$ from Section 1; fixed supports have $d_C = 0$ and $b = f_F$. After solving, the forces exerted by supports, called reactions, are

$$r_C = K_{CF}d_F + K_{CC}d_C - f_C, \tag{7}$$

using the original loads, including any applied at supports. Applied forces and reactions together must balance total force and moment.

The construction also explains why supports are essential. Since D is symmetric and positive definite, K is symmetric and

$$d^T K d = \sum_e t S_e (B_e d_e)^T D (B_e d_e) \geq 0. \tag{8}$$

Half this quantity is the stored elastic energy. An unsupported piece can slide or rotate without straining, giving zero energy and making K singular. If the supports prevent both translations and the rotation of every connected piece, the energy is positive for every nonzero free displacement. Thus A is symmetric positive definite.

How strongly the displacement reacts to small load changes is measured by the condition number

$$\kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}, \tag{9}$$

the ratio of its largest and smallest eigenvalues. A large ratio means a small percentage change in load can cause a much larger percentage change in displacement. These properties guide the solver choices and accuracy checks in the next section.

References

[1] Jeremy Bleyer, *Isotropic and orthotropic plane stress elasticity*, Numerical Tours of Computational Mechanics with FEniCSx, `https://bleyerj.github.io/comet-fenicsx/tours/linear_problems/isotropic_orthotropic_elasticity/isotropic_orthotropic_elasticity.html`. Accessed 10 November 2026.

[2] Jonathan Richard Shewchuk, *Triangle: Generating Delaunay triangulations*, Carnegie Mellon University, Triangle documentation, `https://www.cs.cmu.edu/~quake/triangle.delaunay.html`. Accessed 10 November 2026.

[3] University of Alberta, *Two dimensional solid elements: Triangular elements*, Introduction to Solid Mechanics, Engineering at Alberta Courses, Open Educational Resources, `https://engcourses-uofa.ca/books/introduction-to-solid-mechanics/finite-element-analysis/two-dimensional-solid-elements/triangular-elements/`. Accessed 10 November 2026.